SANKUR KUNDU

Software Engineer | Backend, Systems & Full-Stack

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SUMMARY

Software engineer (B.Tech CSE, 2026) who builds and verifies systems end to end: a fault-tolerant LLM cluster in C++/CUDA, RL post-training (GSM8K **48.1% → 52.5%**), CUDA screening of all **32,593** orbiting objects in **28 s**. Co-founder of Gigzen.

PROJECTS

relay — C++, CUDA and Swift: a disaggregated, fault-tolerant LLM inference cluster

- Built C++/CUDA prefill and decode workers on custom flash-attention and flash-decoding kernels, a KV-cache transfer engine that streams each layer while the next computes, and a Swift control plane with prefix-aware routing.
- On a T4, cut 2,048-token prefill **8.7×** (3.7 s → 0.43 s) and batch-32 decode **5.4×**; streaming hides **99–100%** of the KV transfer. Output matches Hugging Face and stays bit-identical when batched, cached or split across workers.
- Fault-tolerant: watchdogs catch killed, frozen or stalled workers and their requests resume elsewhere with identical output (**384/384** through 60 kills; no request lost in **300** fault-injection seeds).

kessler — C++/CUDA and Python: an independent auditor of space traffic

- Built SGP4 and all-vs-all conjunction screening in C++/CUDA: all **32,593** tracked objects, every pair, a day ahead in **28 s** on a T4; matches CelesTrak SOCRATES to **0.24 m** (median) and NASA CARA to **1.6e-8**.
- Audited SpaceX's **207,152** reported avoidance maneuvers from public data: only about **half** are visible, and satellites that maneuvered had predicted close approaches just **1.3–1.5×** as often.

quorumdb — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built a distributed SQL database that psql connects to: LSM storage, Raft and self-rebalancing multi-Raft, serializable ACID transactions (Percolator) and a SQL planner; **1,500 tpmC** on a TPC-C-derived benchmark.
- Verified by deterministic simulation (**10,000** Raft clusters, 610K crashes, histories machine-checked) and a TLA+ model (337K states): found **6 real bugs** and catches all 15 planted ones.

lipi — Python and PyTorch: cutting the token tax on Indian languages

- Measured 24 Indian languages on 15 production tokenizers and extended Qwen 2.5's: fewer tokens than **GPT-4o's and Gemini's** in **23 of 24** languages; trained into Qwen2.5-0.5B on 2× T4 (DDP, ZeRO-1), Odia trains **5.4× faster**.

Additional projects

- **ratchet**: RL post-training (GRPO) from scratch on a custom C++/CUDA engine: Qwen2.5-0.5B GSM8K **48.1% → 52.5%**.
- **crucible**: sandboxes for agent RL forked in **4.5 ms**; GRPO graded in them lifted MBPP **34.9% → 38.6%** (3 seeds, p = 0.013).
- **switchyard**: MoE serving across 2 GPUs, **2.2×** Hugging Face; **tandem**: distributed training **bit-identical** to PyTorch.

OPEN SOURCE

- vLLM **#60955** (under review): found its FlashInfer prefill tests compared non-causal attention with a causal reference; with my fix, **32 of 32** cases pass on a T4 and **all 32** fail if the bug returns.
- Meta ExecuTorch **#23677** (under review): fixed out-of-bounds reads in **2-bit/4-bit** embedding kernels for on-device LLMs.
- PyTorch **#199875**, **#199441**, **#199646** (under review): torch.compile determinism fix; **13** skips removed, **8** tests enabled.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer **Apr 2026 – Present**

Early-stage startup (incorporation in progress) · Sole engineer across every product

- Sole engineer on **Waggle**, three Android apps on Supabase PostgreSQL: **53 tables, 90 RLS policies, 154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: first-accept-wins dispatch, 4-digit handover codes, Razorpay/UPI payments reconciled by UTR, GST invoices and pg_cron billing.
- Security-tested it with **150 automated attacks** on its database access rules (**all blocked**) and a 42-step Playwright run.
- Built **Populace**, an open-source multi-user testing tool (**npm** CLI, GitHub Action, Electron app); its first run found **5 defects in 3.5 minutes** after manual testing had passed, one breaking every signup.

SKILLS

CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks
Languages: C++, Rust, Python, Swift (C++ interop), TypeScript, JavaScript, SQL (PL/pgSQL), Java
Backend & Data: PostgreSQL, Supabase, Row-Level Security, pg_cron, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI
Frontend & Mobile: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS
Systems & performance: CUDA (tensor cores), LLM serving, MoE expert parallelism, RL post-training, DDP/ZeRO, SIMD
Testing, DevOps & Cloud: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata **Sep 2022 – Apr 2026**